

Network Device Test

1. Behaviour of HUB :

>> A hub is a basic network device used to connect multiple computers in a LAN. It works at Layer 1 (Physical Layer) of the OSI model.

>> Broadcasts data to all ports. It is not intelligent and does not learn mac address.

2. Behaviour of Switch :

>> A switch is a network device that works mainly at layer 2 (Data Link) of OSI model.

>> Learns the source mac address and stores it in its mac address table.

>> Learns mac addresses and forwards data intelligently.

3. Behaviour of Router :

>> A router is a network device that works mainly at layer 3 (Network layer) of OSI model. Its main job is to connect different networks and forward packets between them.

>> The router checks PC₁'s destination IP address, finds the appropriate route and forwards the packet toward Network B.

>> Checks destination IP + selects the best route + forwards the packet to another network = Router.